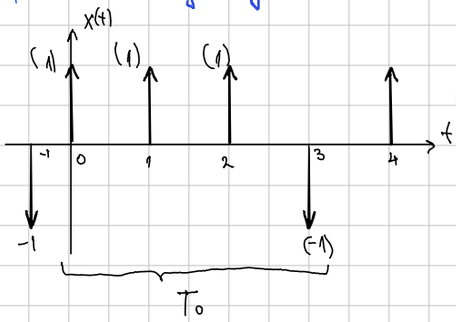
4. TJEDAN

~ Fourierov red ~

Zadatak 1.) Kontinuirani signal $x(t)$ periodičan je s periodom $T=4s$. Prikažite ovaj signal Fourierovim redom, te odredite koeficijente tog reda.



CTFS

$F_k = ?$

$$X_k = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-j\omega_0 k t}\, dt$$

$$\rightarrow x(t) = \delta(t) + \delta(t-1) + \delta(t-2) + \delta(t-3)$$

$$X_k = \frac{1}{4}\int_{-0.5}^{3.5} x(t)\, e^{-j\omega_0 k t}\, dt \qquad \omega_0 = \frac{2\pi}{T} = \frac{\pi}{2}$$

$$X_k = \frac{1}{4}\left[\int_{-0.5}^{0.5} \delta(t)\cdot e^{-j\frac{\pi}{2}kt}\, dt + \int_{0.5}^{1.5} \delta(t-1)\, e^{-j\frac{\pi}{2}kt}\, dt + \int_{1.5}^{2.5} \delta(t-2)\, e^{-j\frac{\pi}{2}kt}\, dt + \int_{2.5}^{3.5} \delta(t-3)\, e^{-j\frac{\pi}{2}kt}\, dt\right]$$

$$= \frac{1}{4}\left[e^{-j\frac{\pi}{2}k\cdot 0}\underbrace{\int_{-0.5}^{0.5}\delta(t)\, dt}_{1} + e^{-j\frac{\pi}{2}k}\underbrace{\int_{0.5}^{1.5}\delta(t-1)\, dt}_{1} + e^{-j\pi k}\underbrace{\int_{1.5}^{2.5}\delta(t-2)\, dt}_{1} + e^{-j\frac{3\pi}{2}k}\underbrace{\int_{2.5}^{3.5}\delta(t-3)\, dt}_{1}\right]$$

$$= \frac{1}{4}\left[1 + e^{-j\frac{\pi}{2}k} + e^{-j\pi k} + e^{-j\frac{3\pi}{2}k}\right]$$

$$e^{-j\frac{\pi}{2}k} = \left(e^{-j\frac{\pi}{2}}\right)^k = \left(\cos\left(\frac{\pi}{2}\right) - j\sin\left(\frac{\pi}{2}\right)\right)^k$$

$$e^{-j\frac{\pi}{2}k} = (-j)^k$$

$$e^{-j\pi k} = \left(e^{-j\pi}\right)^k = \left(\cos\pi - j\sin\pi\right)^k$$

$$e^{-j\pi k} = (-1)^k$$

$$e^{-j\frac{3\pi}{2}k} = \left(e^{-j\frac{3\pi}{2}}\right)^k = \left(\cos\frac{3\pi}{2} - j\sin\frac{3\pi}{2}\right)^k$$

$$e^{-j\frac{3\pi}{2}k} = j^k$$

$$\Rightarrow \boxed{X_k = \frac{1}{4}\left[1 + (-j)^k + (-1)^k - j^k\right]}$$

Amplitudni spektar:

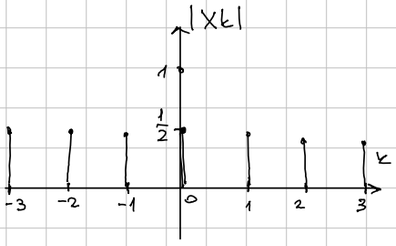
$$|X_k| = \frac{1}{4}\left[1 + (-j)^k + (-1)^k - j^k\right]$$

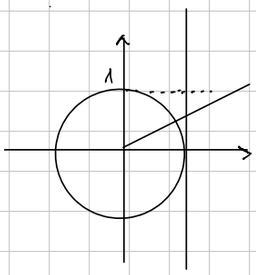
$$X_0 = \frac{1}{4}\left[1 + 1 + 1 - 1\right] = \frac{1}{2} \rightarrow |X_0| = \frac{1}{2}$$

$$X_1 = \frac{1}{4}\left[1 + (-j) + (-1) - j\right] = \frac{-2j}{4} \rightarrow |X_1| = \frac{1}{2}$$

$$X_2 = \frac{1}{4}\left[\cancel{1} + (\cancel{-1}) + 1 - (-1)\right] = \frac{1}{2} \rightarrow |X_2| = \frac{1}{2}$$

$$X_3 = \frac{1}{4}\cdot 2j \longrightarrow |X_3| = \frac{1}{2}$$





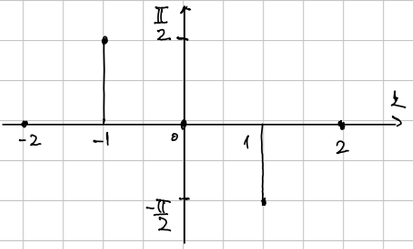
Fazni spektar:

$$\measuredangle X_0 = \operatorname{arctg}\frac{y}{x} = \operatorname{arctg}\frac{0}{\frac{1}{2}} = \emptyset \rightarrow \measuredangle X_0 = 0$$

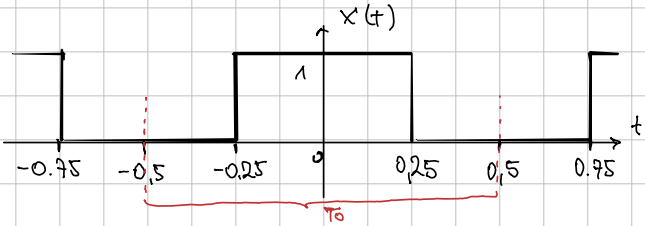
$$\measuredangle X_1 = \operatorname{arctg}\frac{-\frac{1}{2}}{0} \longrightarrow \measuredangle X_1 = \frac{-\pi}{2}$$

$$\measuredangle X_2 = \operatorname{arctg}\frac{0}{\frac{1}{2}} \longrightarrow \measuredangle X_2 = 0$$

$$\measuredangle X_3 = \operatorname{arctg}\frac{\frac{1}{2}}{0} \longrightarrow \measuredangle X_3 = \frac{\pi}{2}$$



Zadatak 2.) Zadan je periodičan signal $x(t)$. Odredite srednju snagu ovog signala (u vremenskoj i frekvencijskoj domeni), te aproksimirajte signal Fourierovim redom.



$P = ?$ u vremenskoj i frekvencijskoj domeni

$$\hat{f}(t) = \sum F_k \cdot \psi_k$$

$$X_k = \frac{1}{T_0}\int_{T_0} x(t)\cdot e^{-j\omega_0 k t}\,dt$$

$$\underbrace{P = \frac{1}{T_0}\int_{T_0} |x(t)|^2\,dt}_{\text{vremenska domena}} = \underbrace{\sum_{k=-\infty}^{\infty} |X_k|^2}_{\text{frekvencijska domena}}$$

$$X_k = \frac{1}{1}\int_{-0.5}^{0.5} x(t)\, e^{-j\frac{2\pi}{1}kt}\,dt$$

$$X_k = \int_{-0.25}^{0.25} 1\cdot e^{-j2\pi k t}\,dt = \frac{1}{-j2\pi k}\, e^{-j2\pi k t}\Big|_{-0.25}^{0.25} = \frac{1}{-j2\pi k}\underbrace{\left(e^{-j\frac{\pi}{2}k} - e^{+j\frac{\pi}{2}k}\right)}_{(-2j)\sin\left(\frac{\pi}{2}k\right)}$$

$$X_k = \frac{1}{\cancel{-j}2\pi k}\,\cancel{(-2j)}\,\sin\left(\frac{\pi}{2}k\right) = \frac{1}{2}\cdot\frac{\sin\left(\frac{\pi}{2}k\right)}{\frac{\pi}{2}k} = \underbrace{\boxed{\frac{1}{2}\operatorname{sinc}\left(\frac{\pi}{2}k\right)}}_{F_k}$$

* poseban slučaj: $k=0$

$$X_0 = \frac{1}{1}\int_{-0.25}^{0.25} 1\cdot e^{-j2\pi 0 t}\,dt = 0{,}5 \longrightarrow \boxed{F_0 = \frac{1}{2}}$$

Srednja snaga u vremenskoj domeni $\Rightarrow P = \frac{1}{1}\int_{-0.5}^{0.5} |x(t)|^2\,dt$

$$P = \int_{-0.25}^{0.25} 1\,dt = 0.25 + 0{,}25 \longrightarrow \boxed{P(t) = \frac{1}{2}}$$

Srednja snaga u frekvencijskoj domeni $\Rightarrow P = \sum_{k=-\infty}^{\infty} |X_k|^2$

* budući da je u $k=0$ poseban slučaj, odvojimo $\rightarrow P = \sum_{k=-\infty}^{-1} |X_k|^2 + |X_0|^2 + \sum_{k=1}^{\infty} |X_k|^2$

$$P = 2\sum_{k=1}^{\infty} |X_k|^2 + |X_0|^2$$

$$P = 2\sum_{k=1}^{\infty}\left(\frac{1}{2}\cdot\frac{\sin\left(\frac{\pi}{2}k\right)}{\frac{\pi}{2}k}\right)^2 + \frac{1}{4} = \frac{1}{2}\sum_{k=1}^{\infty}\frac{\sin^2\left(\frac{\pi}{2}k\right)}{\left(\frac{\pi}{2}k\right)^2} + \frac{1}{4} \rightarrow \boxed{P \approx \frac{1}{2}}$$

Zadatak 3.) Odredite rastav u Fourierov red signala

$$x(t) = 10\cos(50\pi t) + 5\sin(100\pi t) + \sin\left(150\pi + \frac{2\pi}{3}\right) + \cos\left(200\pi t + \frac{\pi}{4}\right)$$

te skicirajte dobiveni amplitudni i fazni spektar. Ako signal otipkamo s periodom otipkavanja $T_s = 0{,}02$, je li došlo do preklapanja spektra?

signal je periodičan $\rightarrow \omega_1 = 50\pi$

$10\cos(50\pi t) \rightarrow 50\cancel{\pi} = \frac{2\cancel{\pi}}{T_1} \rightarrow T_1 = 0{,}04\,s = \frac{1}{25}\,s$

$5\cos(100\pi t) \rightarrow 100\cancel{\pi} = \frac{2\cancel{\pi}}{T_2} \rightarrow T_2 = 0.02\,s = \frac{1}{50}\,s$

$\sin\left(150\pi + \frac{2\pi}{3}\right) \rightarrow 150\pi = \frac{2\pi}{T_3} \rightarrow T_3 = 0{,}0067 = \frac{1}{75}\,s$

$\cos\left(200\pi + \frac{\pi}{4}\right) \rightarrow 200\pi = \frac{2\pi}{T_3} \rightarrow T_4 = 0{,}001 = \frac{1}{100}\,s$

Ukupni period je najmanji zajednički djelitelj $\Rightarrow \frac{1}{25}\,s$

$T_0 = \frac{1}{25}\,s \qquad T_2 = \frac{T_0}{2} \qquad T_3 = \frac{T_0}{3} \qquad T_4 = \frac{T_0}{4}$ * zadano $T_s = \frac{1}{50}\,s$

$\Downarrow$

$\omega_0 = \frac{2\pi}{T_0} \Rightarrow \omega_0 = 50\pi\ rad/s \qquad \omega_2 = 2\omega_0 \qquad \omega_3 = 3\omega_0 \qquad \omega_4 = 4\omega_0$

$$x(t) = 10\cdot\frac{1}{2}\left(e^{j\omega_0 t} + e^{-j\omega_0 t}\right) + 5\cdot\frac{1}{2j}\left(e^{j2\omega_0 t} - e^{-j2\omega_0 t}\right) + \frac{1}{2j}\left(e^{j\left(3\omega_0 t + \frac{2\pi}{3}\right)} - e^{-j\left(3\omega_0 t + \frac{2\pi}{3}\right)}\right)$$
$$+ \frac{1}{2}\left(e^{j\left(4\omega_0 t + \frac{\pi}{4}\right)} + e^{-j\left(4\omega_0 t + \frac{\pi}{4}\right)}\right)$$

Budući da je frekvencija očitavanja $f_s = 50\,Hz$, a $x(t)$ se sastoji od signala frekvencija: 25, 50, 75 i 100 Hz,

da ne bi došlo do preklapanja $f_s >$ od najveće frekv signala $\times 2$

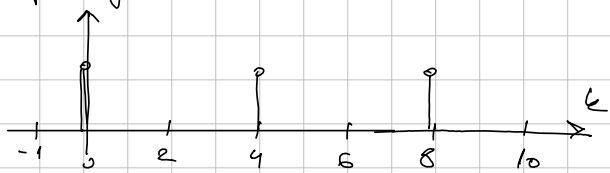
(jer $T_s < 2T_{max}$) $\longrightarrow f_s > 200\,Hz$ što ne vrijedi

$\Longrightarrow$ ALIASING

Zadatak 4.) Izračunajte Fourierove koeficijente vremenski kont signala $x(t) = \sum_{k=-\infty}^{\infty} \delta(t-4k)$ ← CTFS

$$X_k = \frac{1}{T_0}\int_{T_0} x(t)\cdot e^{-j\omega_0 kt}\, dt$$

↳ ponavlja se za svaka 4



Znači period je $(-1,3) \to 4$

$$X_k = \frac{1}{4}\int_{-1}^{3} \delta(t-4k)\, e^{-j\frac{2\pi}{T_0}kt}\, dt$$

$$X_k = \frac{1}{4}\int_{-1}^{3} \underbrace{\delta(t\cancel{-4k})}_{\delta(t)}\, e^{-j\frac{\pi}{2}\cancel{k}t}\, dt$$

$k=1$

onda kada je $t=4k$ onda signal daje vrijednost 1 (tj. $\neq 0$)

Kod Fourierovih koef. za CTFS integriramo samo unutar prve periode!

$$X_k = \frac{1}{4}\int_{-1}^{3} \delta(t)\, e^{-j\frac{\pi}{2}t}\, dt = \frac{1}{4}e^{-j\frac{\pi}{2}0} = \boxed{\frac{1}{4}}$$

$t_0 = 0$

Spektar je jednak $\frac{1}{4}$ za svaki k

→ dobiveni spektar je diskretan

Zadatak 5.) Nađite Fourierove transformacije, te amplitudne, fazne, realne i imaginarne spektre sljedećih signala.

Odredite energiju zadanih signala u vremenskoj i u frekvencijskoj domeni. U kakvom su odnosu ove dvije energije?

a) $x(t) = e^{-t}\mu(t) \rightarrow$ CTFT

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt = \int_0^{\infty} e^{-t} e^{-j\omega t}\, dt = \int_0^{\infty} e^{-(1+j\omega)t}\, dt = \frac{-1}{1+j\omega} e^{-(1+j\omega)t}\Big|_0^{\infty}$$

$$X(j\omega) = \frac{-1}{1+j\omega}\Big(\underbrace{\frac{1}{e^{\infty}}}_{0} - 1\Big) = \frac{1}{1+j\omega}\cdot\frac{1-j\omega}{1-j\omega} = \frac{1-j\omega}{1+\omega^2} = X(j\omega)$$

Amplitudni spektar: $X(j\omega) = \dfrac{1-j\omega}{1+\omega^2}$

$$X(j\omega) = \underbrace{\frac{1}{1+\omega^2}}_{Re} - j\underbrace{\frac{\omega}{1+\omega^2}}_{Jm} \longrightarrow |X(j\omega)| = \sqrt{\frac{1}{(1+\omega^2)^2} + \frac{\omega^2}{(1+\omega^2)^2}} = \frac{\sqrt{1+\omega^2}}{(1+\omega^2)} = \frac{1}{\sqrt{1+\omega^2}}$$

$$\frac{\sqrt{1+\omega^2}}{(1+\omega^2)} = |X(j\omega)|$$

Fazni spektar: $\varphi = \operatorname{arctg}\dfrac{Y}{X} \Rightarrow \varphi = \operatorname{arctg}\dfrac{\frac{-\omega}{1+\omega^2}}{\frac{1}{1+\omega^2}}$ $\quad \varphi = \operatorname{arctg}(-\omega)$

Realni $\quad Re = \dfrac{1}{1+\omega^2}$

Imaginarni $\quad Jm = -\dfrac{\omega}{1+\omega^2}$

Energija:

- vremenska domena: $E = \int_{-\infty}^{\infty} |x(t)|^2\, dt = \int_0^{\infty} e^{-2t}\, dt = \frac{-1}{2} e^{-2t}\Big|_0^{\infty} = \boxed{\frac{1}{2}}$

- frekvencijska domena: $E = \dfrac{1}{2\pi}\displaystyle\int_{-\infty}^{\infty} |X(j\omega)|^2\, d\omega$

$$E = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left|\frac{1-j\omega}{1+\omega^2}\right|^2 d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty}\left(\frac{1}{\sqrt{1+\omega^2}}\right)^2 d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{1}{1+\omega^2}\, d\omega$$

$$\int \frac{dx}{a^2x^2+b^2} = \frac{1}{ab}\operatorname{arctg}\left(\frac{ax}{b}\right)$$

$$E = \frac{1}{2\pi}\cdot\frac{1}{1}\operatorname{arctg}\left(\frac{\omega}{1}\right)\Big|_{-\infty}^{\infty} = \frac{1}{2\pi}\left(\operatorname{arctg}(\infty) + \operatorname{arctg}(\infty)\right)$$

$$= \frac{1}{2\pi}\left(\frac{\pi}{2} + \frac{\pi}{2}\right) = \boxed{\frac{1}{2}}$$

b) $x(t) = e^{t}\mu(-t)$

$$X(j\omega)=\int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt = \int_{-\infty}^{0} e^{t} e^{-j\omega t}\, dt = \int_{-\infty}^{0} e^{(1-j\omega)t}\, dt = \frac{1}{1-j\omega}\left(1-\underbrace{e^{-\infty}}_{0}\right)$$

$$\boxed{X(j\omega)=\frac{1+j\omega}{1+\omega^2} = \frac{1}{1+\omega^2} + j\,\frac{\omega}{1+\omega^2}}$$

Amplitudni spektar: $|X(j\omega)| = \sqrt{\frac{1}{(1+\omega^2)^2} + \frac{\omega^2}{(1+\omega^2)^2}} = \frac{\sqrt{1+\omega^2}}{1+\omega^2} = \frac{1}{\sqrt{1+\omega^2}}$

Fazni spektar: $\measuredangle X(j\omega) = \varphi = \operatorname{arctg}\frac{\frac{\omega}{\cancel{1+\omega^2}}}{\frac{1}{\cancel{1+\omega^2}}} = \operatorname{arctg}\omega$

Realni $\quad Re = \frac{1}{1+\omega^2}$

Imaginarni $\quad Im = \frac{\omega}{1+\omega^2}$

ENERGIJA:

- vremenska domena: $E=\int_{-\infty}^{\infty}|x(t)|^2 dt = \int_{-\infty}^{0} e^{2t}\, dt = \frac{1}{2}e^{t}\Big|_{-\infty}^{0} = \boxed{\frac{1}{2}}$
- frekvencijska domena: $E=\frac{1}{2\pi}\int_{-\infty}^{\infty}|X(j\omega)|^2 d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{1}{1+\omega^2}\, d\omega$

$$E = \frac{1}{2\pi}\operatorname{arctg}\left(\frac{\omega}{1}\right)\Big|_{-\infty}^{\infty} = \frac{1}{2\pi}\left(\operatorname{arctg}(\infty)+\operatorname{arctg}(\infty)\right) = \boxed{\frac{1}{2}}$$

$$\int_{-\infty}^{0} e^{t} + \int_{0}^{\infty} e^{-t}$$

c) $x(t) = e^{-|t|} \rightarrow X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\, dt = \cancel{2\int_{0}^{\infty} e^{-t}\cdot e^{-j\omega t}\, dt}$

$$X(j\omega)=2\int_{0}^{\infty} e^{-(1+j\omega)t}\, dt = 2\left(\frac{-1}{1+j\omega}\right)e^{-(1+j\omega)t}\Big|_{0}^{\infty} = \frac{-2}{1+j\omega}(0-1) = \frac{2}{1+j\omega}$$

$$X(j\omega) = 2\left(\frac{1}{1+\omega^2} - j\,\frac{\omega}{1+\omega^2}\right)$$

Amplitudni: $|X(j\omega)| = 2\sqrt{\frac{1}{(1+\omega^2)^2} + \frac{\omega^2}{(1+\omega^2)^2}} = 2\,\frac{1}{\sqrt{1+\omega^2}}$

Fazni spektar: $\measuredangle X(j\omega) = \varphi = \operatorname{arctg}(-\omega) = -\operatorname{arctg}(\omega)$

Realni: $\frac{1}{1+\omega^2}$ $\qquad$ Imaginarni: $\frac{\omega}{1+\omega^2}$

ENERGIJA:

$$E = \int_{-\infty}^{\infty}|x(t)|^2 dt = 2\int_{0}^{\infty} e^{-2t}\, dt = \cancel{2}\left(\frac{-1}{\cancel{2}}\right)e^{-2t}\Big|_{0}^{\infty} = -1\cdot(-1) = 1$$

$$E = \frac{1}{2\pi}\int_{-\infty}^{\infty}|X(j\omega)|^2 d\omega = \frac{1}{\cancel{2}\pi}\int_{-\infty}^{\infty}\left(\frac{2}{\sqrt{1+\omega^2}}\right)^2 d\omega = \frac{2}{\pi}\int_{-\infty}^{\infty}\frac{1}{1+\omega^2}\, d\omega$$

$$E = \frac{2}{\pi}\operatorname{arctg}(\omega)\Big|_{-\infty}^{\infty} = \frac{2}{\pi}\cdot\left(\frac{\pi}{2}+\frac{\pi}{2}\right) = \boxed{2}$$

c) $x(t) = e^{-|t|}$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\,dt = \int_{-\infty}^{0} e^{t} \cdot e^{-j\omega t}\,dt + \int_{0}^{\infty} e^{-t} e^{-j\omega t}\,dt = \int_{-\infty}^{0} e^{(1-j\omega)t}\,dt +$$

$$+ \int_{0}^{\infty} e^{-(1+j\omega)t}\,dt = \frac{1}{1-j\omega} e^{(1-j\omega)t}\Big|_{-\infty}^{0} - \frac{1}{1+j\omega} e^{-(1+j\omega)t}\Big|_{0}^{\infty}$$

$$= \frac{1}{1-j\omega} + \frac{1}{1+j\omega} = \frac{\cancel{1+j\omega} + \cancel{1-j\omega}}{1+\omega^2} = \boxed{\frac{2}{1+\omega^2}}$$

Amplitudni spektar: $|X(j\omega)| = \dfrac{2}{1+\omega^2}$ Realni $\dfrac{2}{1+\omega^2}$

Fazni spektar $\measuredangle X(j\omega) = \emptyset$ Imaginarni $\emptyset$

ENERGIJA:

$$E = \int_{-\infty}^{\infty} |x(t)|^2\,dt = \int_{-\infty}^{0} e^{2t}\,dt + \int_{0}^{\infty} e^{-2t}\,dt = \frac{1}{2} - \frac{1}{2}\cdot(-1) = \boxed{1}$$ vremenska domena

$$E = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2\,d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{4}{(1+\omega^2)^2}\,d\omega = \frac{2}{\pi}\left(\frac{\omega}{2(1+\omega^2)} + \frac{1}{2}\operatorname{arctg}(\omega)\right)\Bigg|_{-\infty}^{\infty}$$

$$= \frac{\cancel{2}}{\pi}\left(\cancel{\frac{\frac{1}{\omega}}{\cancel{2}\left(\frac{1}{\omega^2}+1\right)}}^{\,=0} + \frac{1}{\cancel{2}}\operatorname{arctg}(\omega)\right)\Bigg|_{-\infty}^{\infty} = \frac{1}{\pi}\left(\operatorname{arctg}(\infty) + \operatorname{arctg}(\infty)\right) = \boxed{1}$$ frekvencijska domena

Zadatak 6.) $x(t) = e^{2t}\mu(-t)$

a) Nađite Fourierovu transf. zadanog signala

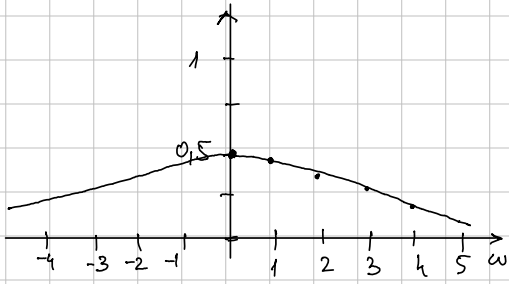
$$X(j\omega) = \int_{-\infty}^{\infty} e^{2t}\mu(-t)e^{-j\omega t}\,dt = \int_{-\infty}^{0} e^{2t}\cdot e^{-j\omega t}\,dt = \int_{-\infty}^{0} e^{(2-j\omega)t}\,dt = \frac{1}{2-j\omega}e^{(2-j\omega)t}\Big|_{-\infty}^{0}$$

$$X(j\omega) = \frac{1}{2-j\omega}\left(1-0\right) = \frac{1}{2-j\omega}\cdot\frac{2+j\omega}{2+j\omega} = \frac{2+j\omega}{4+\omega^2} = \frac{2}{4+\omega^2} + j\frac{\omega}{4+\omega^2}$$

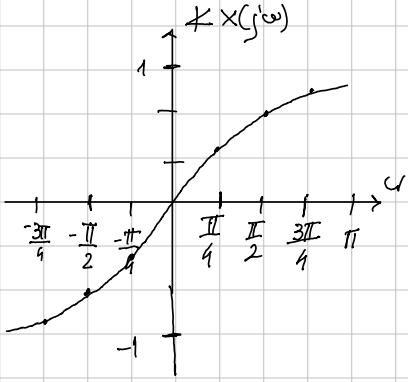
b) Nacrtaj amplitudni i fazni spektar:

Amplitudni:

$$|X(j\omega)| = \sqrt{\frac{4}{(4+\omega^2)} + \frac{\omega^2}{(4+\omega^2)^2}} = \frac{1}{\sqrt{4+\omega^2}}$$



Fazni spektar: $\measuredangle X(j\omega) = \operatorname{arctg}\left(\frac{\omega}{2}\right)$



c) Energija signala u vremenskoj domeni

$$E = \int_{-\infty}^{\infty} |x(t)|^2\,dt = \int_{-\infty}^{0} e^{4t}\,dt$$

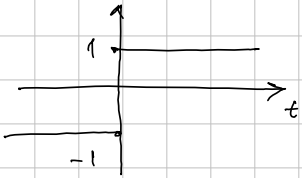
$$E = \frac{1}{4}e^{4t}\Big|_{-\infty}^{0} = \boxed{\frac{1}{4}}$$

d) Energija u frekv. domeni → Parseval!

$$E = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2\,d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{1}{4+\omega^2}\,d\omega = \frac{1}{2\pi}\cdot\frac{1}{2}\operatorname{arctg}\left(\frac{\omega}{2}\right)\Big|_{-\infty}^{\infty} = \frac{1}{4\pi}\operatorname{arctg}\left(\frac{\omega}{2}\right)\Big|$$

$$\boxed{E = \frac{1}{4}}$$

Zadatak 7.) Nađite FT od sgn funkcije: $\mathrm{sgn}(t) = \begin{cases} 1, & t>0 \\ -1, & t<0 \end{cases}$



$$\mathrm{sgn}(t) = -1 + 2\mu(t)$$

$$\cancel{X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} (-1 + 2\mu(t))\, e^{-j\omega t}\, dt = -\int_{-\infty}^{\infty} e^{-j\omega t}\, dt + 2\int_{0}^{\infty} e^{-j\omega t}\, dt}$$

→ najlakše direktnim očitavanjem iz tablica transf: 3. str

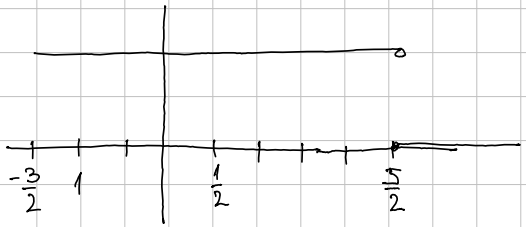
$1 \rightarrow 2\pi\delta(\omega)$

$\mu(t) \rightarrow \pi\delta(\omega) + \frac{1}{j\omega}$

$$\mathrm{sgn}(j\omega) = \underbrace{-2\pi\delta(\omega)}_{=0} + \underbrace{2\left(\pi\delta(\omega) + \frac{1}{j\omega}\right)}_{=0}$$

$$\boxed{\mathrm{sgn}(j\omega) = \frac{2}{j\omega}}$$

Zadatak 8.) FT signala $x(t) = \begin{cases} 1, & -\frac{3}{2} < t < \frac{5}{2} \\ 0, & \text{inače} \end{cases}$



$$X(j\omega) = \int_{-\frac{3}{2}}^{\frac{5}{2}} 1 \cdot e^{-j\omega t}\, dt =$$

$$X(j\omega) = \frac{-1}{j\omega} e^{-j\omega t} \Big|_{-\frac{3}{2}}^{\frac{5}{2}} = \frac{-1}{j\omega}\left(e^{-j\omega\cdot\frac{5}{2}} - e^{j\omega\cdot\frac{3}{2}}\right)$$

$$X(j\omega) = \frac{-1}{j\omega} e^{-j\omega\cdot\frac{1}{2}}\left(e^{-j\omega\frac{4}{2}} - e^{j\omega\frac{4}{2}}\right) = \frac{e^{-j\omega\cdot\frac{1}{2}}}{\cancel{j}\omega}(\cancel{-}2\cancel{j})\sin(2\omega) = \boxed{2\, e^{-j\omega\frac{1}{2}} \frac{\sin(2\omega)}{2\omega}}$$

Zad 9.) FT: $x(t) = 220 \sin\left(50\pi t + \frac{\pi}{3}\right)$

Amplitudni? Fazni? $P = ?$

$x(t)$ je periodičan jer je sinus periodičan $\Rightarrow$ CTFS

$$F_T \Rightarrow X_k = \frac{1}{T_0}\int_{T_0} x(t)\, e^{-j\omega_0 k t}\, dt \qquad \omega = 50\pi \rightarrow T = \frac{2\pi}{\omega} = \frac{1}{25}$$

$$X_k = 25 \int_0^{0{,}04} 220 \sin\left(50\pi t + \frac{\pi}{3}\right) \cdot e^{-j\cdot 50\pi k t}\, dt$$

$$X_k = 5500 \int_0^{0{,}04} \frac{e^{j(50\pi t + \frac{\pi}{3})} - e^{-j(50\pi t + \frac{\pi}{3})}}{-2j} \cdot e^{-j50\pi t} \overset{k=1}{=} \frac{2750}{-j}\int_0^{0{,}04} \left(e^{j50\pi} \cdot e^{j\frac{\pi}{3}} - e^{-j50\pi t} \cdot e^{-j\frac{\pi}{3}}\right)$$

$$X_k = 2750j \int_0^{0{,}04} e^{j\frac{\pi}{3}} - e^{-100\pi t} \cdot e^{-j\frac{\pi}{3}}$$

$$x(t) = 220 \frac{e^{j(50\pi t + \frac{\pi}{3})} - e^{-j(50\pi t + \frac{\pi}{3})}}{2j} = 100 \cdot (-j)\left(e^{j50\pi t} \cdot e^{j\frac{\pi}{3}} - e^{-j50\pi t} e^{-j\frac{\pi}{3}}\right)$$

$$x(t) = 110 \cdot e^{-j\frac{\pi}{2}}\left(e^{j50\pi t} \cdot e^{j\frac{\pi}{3}} - e^{-j50\pi t} \cdot e^{-j\frac{\pi}{3}}\right)$$

Sjeti se kako izgleda kolt: $\sum_{k}^{\infty} \underbrace{X_k}_{\text{ovo tražimo, kompleksni brojevi!}} \cdot e^{j\omega_0 k t}$

$$X_1 = 110 \cdot e^{-j\frac{\pi}{2}} \cdot e^{j\frac{\pi}{3}} = 110\, e^{-j\frac{\pi}{6}} \rightarrow |X_1| = 110, \quad \measuredangle X_1 = -\frac{\pi}{6}$$

$$X_2 = 110 \cdot e^{-j\frac{\pi}{2}} \cdot e^{-j\frac{\pi}{3}} = 110 \cdot e^{-j\frac{5}{6}\pi} \qquad |X_2| = 110 \quad \measuredangle X_2 = \frac{-5}{6}\pi$$

Snaga: $P = \frac{1}{T_0}\int_{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{+\infty} |X_k|^2 \Rightarrow 110^2 + 110^2 = \boxed{24200}$